

Víctor Prim Romero

Backend Developer · Applied AI

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Second-year software development student focused on backend and applied AI. I build software end to end: Python APIs, RAG systems, tool-using agents and Linux deployment with Docker. My main project is a multi-tenant RAG chatbot in production that answers questions about me on victorprim.com, with automated evaluation, cost controls and latency tracking. I have also orchestrated AI with n8n, Dify and Make during a team internship and in personal projects.

SELECTED PROJECTS AND INTERNSHIP

Multi-tenant RAG chatbot: portfolio assistant (personal project, in production)

2026 · [Live demo on victorprim.com](#) · [Case study \(Spanish\)](#)

- FastAPI backend with SSE streaming and a tool-using agent that captures leads and hands off to a human. Multi-tenant, with isolation between tenants verified by tests.
- Retrieval on PostgreSQL with pgvector (HNSW index) and bge-m3 embeddings served locally with Ollama. I switched embedding models after measuring that the previous one discriminated poorly in Spanish, raising the eval from 11/20 to 17/20. Rank-based fusion of two queries fixed short questions, from 10/12 to 12/12.
- Automated per-tenant evaluation: a retrieval eval (19/20) and a conversational eval with permanent regression cases taken from real failures. Returns a non-zero exit code on any failure.
- Cost controls: daily token budget per tenant, per-IP cap, rate limiting behind a reverse proxy, and limits on turns and output tokens.
- Per-stage latency instrumentation stored in the database. It showed that reloading the embedding model added about 5 s to time to first token.
- Deployed on a Linux VPS with Docker Compose and Caddy (HTTPS), with Supabase cloud as the database. Dependency-free JavaScript widget using Shadow DOM, embeddable in any website with a single script tag. 194 tests with pytest.

Stack: Python · FastAPI · PostgreSQL + pgvector (Supabase) · Ollama · Anthropic API · Groq · Docker · Caddy · pytest · JavaScript

AI conversational assistant integrated into an app (internship)

March – September 2026 · Team of 6 · Role: backend, orchestration and infrastructure

- Designed the n8n and Dify workflows connecting the Flutter app to the AI services, including the data contract between the app and the workflows.
- Built a scoped agent with system prompts and RAG to reduce out-of-domain answers.
- Integrated Google Calendar via OAuth to read and create events.
- Set up a shared local server so the whole team tested the same workflow, unifying the environment and freeing RAM on their laptops.
- Defined error handling for the workflows and worked with Git as a team (branches and merges).

Stack: n8n · Dify · Groq · Flutter · Google Calendar OAuth · Docker

Banking system: client app, admin panel and AI chatbot (personal project)

2026 · [Repo](#)

- Two native Android apps (Kotlin and Jetpack Compose) on a shared database, with privilege separation between the admin panel and the client app.
- Supabase database with Row Level Security and roles over sensitive data. Peer-to-peer transfers with history.
- Finance chatbot that answers natural-language questions about the user's own data, run by an n8n agent exposed through a webhook and decoupled from the app.

Stack: Kotlin · Jetpack Compose · Supabase (PostgreSQL, RLS) · n8n · Groq · Ktor

Telegram bot for voice-based appointment scheduling (personal project)

2026 · [Repo](#)

- Audio-to-appointment pipeline: Make.com receives the voice message, and Dify transcribes it and extracts title, date and time with system prompts.
- Creates the Google Calendar event via OAuth and confirms to the user. Workflows are exported without credentials and can be re-imported.

Stack: Telegram Bot API · Make.com · Dify · Gemini · Google Calendar API

Flag Challenge: flag quiz web game (personal project, in production)

2026 · [Repo](#) · [Demo](#)

- React and Vite SPA deployed on Vercel, with three game modes and speed-based scoring.
- Supabase backend: profiles, a real-time leaderboard with Realtime, and RLS policies that include guest mode.

Stack: *React · Vite · Supabase (Auth, RLS, Realtime) · Vercel*

Fitness and nutrition app (personal project, in development)

2026 · [Repo](#)

- Flutter app with Clean Architecture and Riverpod, on Supabase with RLS. AI-based calorie estimation from meal photos is planned.

Stack: *Flutter · Dart · Riverpod · Supabase*

WORK EXPERIENCE

Technical Support Operator, Securitas Direct

June 2024 – September 2025

- Remote fault diagnosis, remote system maintenance and resolution of technical incidents.

Croupier, Casino de Barcelona (Grup Peralada)

March 2022 – March 2023

- Accurate handling of payments and customer service under pressure.

EDUCATION

Higher Vocational Diploma in Multiplatform Application Development (CFGS DAM), Davante MEDAC · in progress (2nd year)

Upper Secondary Education (Bachillerato), Technology track, IES de Sales · 2016 – 2020

TECHNICAL SKILLS

Backend: Python, FastAPI, SQL, PostgreSQL, pgvector, Supabase (RLS, Auth, Realtime), MongoDB, pytest.

Applied AI: RAG, embeddings, evaluation of LLM systems, tool-using agents, system prompts, Anthropic API, Groq, Ollama.

Automation: n8n, Dify, Make.com, webhooks, OAuth.

Infrastructure: Linux (Ubuntu), Docker and Docker Compose, Caddy, Cloudflare Workers, Vercel, Git and GitHub.

Other languages and mobile: Kotlin (Jetpack Compose), Dart (Flutter, Riverpod), JavaScript, TypeScript, React, C#.

AI-assisted development: Claude Code with specs and skills.

LANGUAGES

Spanish and Catalan (native) · English (advanced)